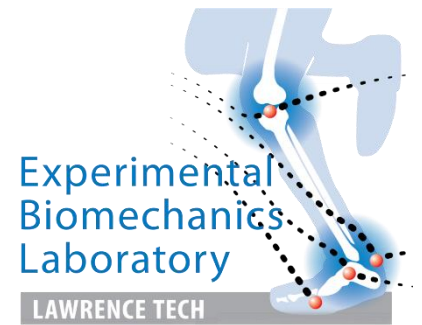




Eric G. Meyer, PhD

[Associate Professor, Biomedical Engineering](#)
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Dr. Meyer specializes in human biomechanics research and developing wearable technology innovations. Meyer received his PhD in Mechanical Engineering (Biomechanics) from Michigan State University. The primary topic of his research was lower extremity biomechanics during sports injury. He completed his post-doc at the Centre for Bioengineering at Trinity College Dublin, Ireland in cartilage tissue engineering and mechanobiology. He was an invited speaker in the session “Solving the mystery of non-contact ACL injuries - how do they happen?” at the 2011 IOC World Conference on Prevention of Injury & Illness in Sport, in Monte-Carlo, Monaco. Meyer directs the LTU Experimental Biomechanics Laboratory, which combines in vivo testing of human subjects with cadaver and surrogate in vitro models and computer simulations to understand normal health, acute injury and degenerative mechanisms. He leads Wearable Technology Innovation Center (WTIC) collaborations that use sensing, perception, actuation and control in the development of next generation robotics, haptics, quantified-self and protective devices. Meyer has published a book chapter and 44 peer-reviewed journal and conference proceeding articles that have received more than 2750 peer citations. He has given more than 100 conference, workshop and invited presentations, including as the Presidential Colloquium speaker for the 2023 LTU Research Day.

PROFESSIONAL EXPERIENCE

Lawrence Technological University, Southfield, Michigan

- Associate Professor, [Biomedical Engineering Department](#) (2016-present)
- Director, Wearable Technology Innovation Center (2019-present)
- Assistant Professor, Biomedical Engineering Program (2010-2016)
- Director, Master of Science Program in Biomedical Engineering (2014-2019)
- Director, Experimental Biomechanics Laboratory (2010-present)

Trinity College Dublin, Dublin, Ireland (2009-2010)

- Post-doctoral Research Fellow, [Centre for Bioengineering](#), Soft Tissue Mechanobiology Research Group
- Directed by Dr. Daniel J. Kelly, Chair of Tissue Engineering

EDUCATION

Michigan State University, East Lansing, Michigan

Ph.D. Mechanical Engineering (2008)

- Dissertation: “[Biomechanical response of the lower extremity to injury level forces in sports loading scenarios](#)”:
- Faculty Advisor: Dr. Roger C. Haut, University Distinguished Professor

M.S. Engineering Mechanics (2004)

- Thesis: “[Biomechanical response of tissues to various blunt impact orientations on the knee](#)”

B.S. Engineering Mechanics, Cognate: Biomechanics (2002) GPA: 3.5/4.0

- Study Abroad: Mechanical Eng. Katholieke Universiteit Leuven, Leuven, Belgium

- Senior Research Project: “Effect of intramedullary pin-plate combination on the mechanical properties of pantarsal arthrodesis: A comparative *in vitro* functional analysis in dogs”

TEACHING EXPERIENCE

BME 3113 "Wearable Technology Design" *Newly developed course*
BME 6303 "Cellular Mechanobiology" *Newly developed course*
BME 5303 "Engineering Applications in Orthopedics" *Newly developed course*
BME 4313 "Tissue Mechanics" *Newly developed course*
BME 3303 & 3301 "Introduction to Biomechanics" & Lab *Newly developed courses*
BME 1002 "Introduction to BME"
EGE 1001 "Fund. of Eng. Design Projects" *Newly developed course*
BME 4013 & 4022 "Biomedical Projects 1 & 2"
BME 6503 "Advanced Experimental Methods in BME" *Newly developed course*
BME 6803 & 6903 "Master's Design Project" & "Master's Thesis"
DIS 8613 "PhD Dissertation"

RELATED EXPERIENCE

[Engineering Consulting](#), Okemos, MI (2012-present)

Biomechanics Consultant – Evaluate forensic biomechanics cases of traumatic and chronic injury mechanisms. Design, perform and analyze experimental biomechanical testing.

Orthopaedic Biomechanics Laboratories, MSU (2001-2008)

Research Assistant - Design, perform and analyze biomechanical experiments on bone, ligament and cartilage to investigate orthopaedic, sports medicine and automotive injuries.

Sports Biomechanics, East Lansing, MI (2004-2007)

Consultant - Evaluate biomechanical characteristics of athletic shoes for *Runner's World Magazine* Shoe Guide or directly for shoe companies.

McDonel Hall, MSU (2001-2002)

Residents Hall Mentor - Provide a positive living environment while emphasizing learning, multicultural understanding and ethical decision making in a dormitory of 41 students.

PUBLICATIONS (BOOKS)

1. **Meyer EG**, Haut RC. Biomechanical response of the knee in sports injury scenarios. Chapter in: [Knee Joints: Kinematics, Injury Types and Treatment Options](#). Editor: Mascarenhas R. Nova, 2012 ISBN: 978-1-61942-268-1.
2. **Meyer EG**. Disruptive Biotechnology: Past and Future Promises of Biomedical Engineering and Wearables. LTU Research Day Presidential Colloquium Booklet. 2023 [DOI:10.13140/RG.2.2.21335.24484](https://doi.org/10.13140/RG.2.2.21335.24484).

PUBLICATIONS (PEER-REVIEWED)

Google Scholar statistics: citations > 2890, h-index = 25 (as of 2026)

<http://scholar.google.com/citations?user=3SbYkskAAAAJ>

44. Peponis J, Meyer EG. The Fifth Element of Biomedical Engineering is Innovation: A Quintessential Design Studio Course Focusing on the "Wearable Technology" Ecosystem. [ASEE Annual Conf Proc. 2025](#).
43. Delogu F, Nelson M, Timmons S, Weinstein M, Meyer EG, et al. A Systemic Transformation of an Arts and Sciences Curriculum to Nurture Inclusive Excellence of All Students through Course-Based Research Experiences. [Frontiers in Education 8;10.3389/feduc.2023.1142572:2023](#). *Citations = 7
42. Rossman S, **Meyer EG**, Rundell S. Development of a Finite Element Lumbar Spine Model to Predict Intervertebral Disc Herniation Mechanics. [Comp Methods Biomech 25\(1\):1-13:2022](#).
41. Lewis L, Jaussen P, Scrivener M, Shargel D, Meyer E, Delogu F, Shamir L, Collins S. Frankenstein 200 Years Later: Chemistry, Literature and Pop Culture. [J Chem Education 24;1-4:2019](#).
40. Huwais S, **Meyer EG**. A Novel Osseous Densification Approach in Implant Osteotomy Preparation to Increase Biomechanical Primary Stability, Bone Mineral Density and Bone to Implant Contact. [International J Oral & Maxillofacial Implants 2017;32:27-36](#). *Citations = 429
39. Shimokochi Y, Ambegaonkar JP, **Meyer EG**. Changing sagittal-plane landing styles modulates impact and tibiofemoral force magnitude and directions relative to the tibia. J Athletic Training 2016 [doi: 10.4085/1062-6050-51.10.15](https://doi.org/10.4085/1062-6050-51.10.15) *Citations = 41

38. **Meyer EG**, Ulrey BL. Sensing Angular Kinematics by Embedding an Open Hardware Design Project into a Required Biomechanics Course. [ASEE Annual Conf Proc. 2016.](#)
37. Rust M, Nasir M, **Meyer EG**. Using Quantified Self as a Learning Tool to Engage Students in Entrepreneurially Minded Learning and Engineering Design. [ASEE Annual Conf Proc. 2016.](#)
36. Nasir M, Weaver BT, **Meyer EG**. Project-based Learning in a Forensic Engineering Course. [ASEE Annual Conf Proc. 2016.](#)
35. **Meyer EG**, Nasir M. Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the "Quantified Self" social movement. [ASEE Annual Conf Proc. 2015.](#) *Citations = 11
34. Nasir M, Seta J, **Meyer EG**. Introducing high school students to biomedical engineering through summer camps. [ASEE Annual Conf Proc 2014.](#) *Citations = 14
33. Gerhart A, **Meyer EG**. Combining discipline-specific "Introduction to Engineering" courses into a single multi-discipline course to foster the entrepreneurial mindset with entrepreneurially minded learning. [ASEE Annual Conf Proc 2014](#) *Citations = 45
32. Rundell SA, Weaver B, Guiang A, **Meyer EG**. Characterization of occupant lower extremity behaviour during moderate-to-high speed rear impacts. [SAE Technical Paper. 2013-01-0222.](#)
31. Button K, Wei F, **Meyer EG**, Haut RC. Specimen-specific computational models of ankle sprain produced in a laboratory setting. [J Biomech Eng. 2013;135\(4\):041001.](#) *Citations = 17
30. Shimokochi Y, Ambeganonkar JP, Lee SY, **Meyer EG**, Shultz SJ. Sagittal plane body positions influence risk of non-contact anterior cruciate ligament injuries during a single-leg drop landing. [Knee Surg Sports Trauma Arthroscopy. 2012;21\(4\):888-897.](#) *Citations = 208
29. **Meyer EG**, Wei F, Button K, Haut RC. Determination of ankle ligament strain using a rigid body computational model for sports injury scenarios. [IRCOBI Conf. Proc. 2012;36:277-288.](#) *Citations = 6
28. Wei F, Post JM, Braman JE, **Meyer EG**, Powell JW, Haut RC. Eversion during external rotation of the human cadaver foot produces high ankle sprains. [J Orthop Res. 2012;30\(9\):1423-1429.](#) *Citations = 41
27. Wei F, **Meyer EG**, Braman JE, Powell JW, Haut RC. Rotational stiffness of football shoes influences talus motion during external rotation of the foot. [J Biomech Eng. 2012; 134\(041002\): 1-7.](#) *Citations = 16
26. **Meyer EG**, Buckley CT, Kelly DJ. The influence of construct scale on the functional properties of cartilaginous tissues engineered using bone-marrow derived mesenchymal stem cells. [Tissue Eng. 2012; 18\(3-4\): 382-396.](#) *Citations = 42
25. Vinardell T, Rolfe RA, Buckley CT, **Meyer EG**, Murphy P, Kelly DJ. Hydrostatic pressure and TGF- β 3 interact to regulate the induction and maintenance of a chondrogenic phenotype in joint tissue derived stem cells. [European Cells & Materials J. 2012; 23: 121-134.](#) *Citations = 102
24. **Meyer EG**, Buckley CT, Steward AJ, Kelly DJ. The effect of cyclic hydrostatic pressure on the functional development of cartilaginous tissues engineered using bone marrow derived mesenchymal stem cells. [J Mech Behav Biomed Mats. 2011; 4\(7\): 1257-1265.](#) *Citations = 98
23. Haugh MG, **Meyer EG**, Thorpe SD, Vinardell T, Duffy GP, Kelly DJ. Temporal and spatial changes in cartilage-matrix specific gene expression in mesenchymal stem cells in response to dynamic compression. [Tissue Eng. 2011; 17\(23-24\): 3085-3093.](#) *Citations = 82
22. **Meyer EG**, Baumer TG, Haut RC. Pure passive hyperextension of the human cadaver knee generates simultaneous bi-cruciate ligament rupture. [J Biomech Eng. 2011; 133\(011012\): 1-5.](#) *Citations = 29
21. **Meyer EG**, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension is a more potent promoter of chondrogenic differentiation than dynamic compression. [J Biomech. 2010; 43: 2516-2523.](#) *Citations = 122
20. Wei F, Villwock MR, **Meyer EG**, Powell JW, Haut RC. A biomechanical investigation of ankle injury under external foot rotation in the human cadaver. [J Biomech Eng. 2010; 132\(091001\): 1-4.](#) *Citations = 60
19. **Meyer EG**, Villwock MR, Powell JW, Fouty AJ, Haut RC. Rotational injuries of the lower extremity and the use of an *in situ* risk assessment tool for rotational traction. STARSS Conf Proc 2010.
18. Isaac DI, **Meyer EG**, Kopke KS, Haut RC. Chronic changes in the rabbit tibial plateau following blunt trauma to the constrained tibiofemoral joint. [J Biomech. 2010;43:1682-8.](#) *Citations = 38
17. Isaac DI, **Meyer EG**, Haut RC. Development of a traumatic anterior cruciate ligament and meniscal rupture model with a pilot *in vivo* study. [J Biomech Eng. 2010; 132\(064501\):1-4.](#) *Citations = 27
16. **Meyer EG**, Villwock MR, Haut RC. Osteochondral microdamage from valgus bending of the human knee. [Clin Biomech. 2009;24\(7\):577-582.](#)
15. Villwock MR, **Meyer EG**, Powell JW, Haut RC. Development and evaluation of a surrogate ankle for use with a rotational traction measurement apparatus. [J Sports Eng Tech. 2009;223\(4\):151-158.](#) *Citations = 8

14. Ting D, Cabassu J, Guillou RP, Sinnott MT, **Meyer EG**, Haut RC, Degardin LM. In vitro evaluation of the effect of fracture configuration on the mechanical properties of standard and novel interlocking nail systems in bending. *Vet Surg.* 2009;38(7):881-887. **Citations = 32*
13. Dejardin LM, Guillou RP, Ting D, Sinnott MT, **Meyer EG**, Haut RC. Effect of bending direction on the mechanical behavior of interlocking nail systems. *Vet Comp Orthop Trauma.* 2009;22(4):264-269. **Citations = 37*
12. Villwock MR, **Meyer EG**, Powell JW, Fouty AJ, Haut RC. The effects of various infills and fiber structures on generating rotational traction on an artificial surface. *J Sports Eng Tech.* 2009;223:11-19. **Citations = 45*
11. Villwock MR, **Meyer EG**, Powell JW, Fouty AJ, Haut RC. Football playing surface components and shoe designs may affect lower extremity injury risk potential: An in situ torsional resistance assessment. *Am J Sports Med.* 2009;37(3):518-525. **Citations = 224*
10. **Meyer EG** and Haut RC. Anterior cruciate ligament injury induced by internal tibial torsion of tibiofemoral compression. *J Biomech.* 2008;41:3377-3383. **Citations = 298*
9. Isaac DI, **Meyer EG**, Haut RC. Chondrocyte damage and contact pressures following impact on the rabbit tibiofemoral joint. *J Biomech Eng.* 2008;130(041018):1-5. **Citations = 41*
8. **Meyer EG**, Baumer TG, Slade J, Smith WE, Haut RC. Tibio-femoral contact pressures and osteochondral microtrauma during ACL rupture due to excessive compressive loading and internal torque of the human knee. *Am J Sports Med.* 2008;36:1966-1977. **Citations = 155*
7. Guillou RP, Frank JD, Sinnott MT, **Meyer EG**, Haut RC, Dejardin LM. Biomechanical evaluations of medial plating for pantarsal arthrodesis: Comparative in vitro study in dogs. *Am J Vet Res.* 2008;69(11):1406-1412. **Citations = 13*
6. Hansen M, Long L, Wertheimer S, **Meyer EG**, Haut RC. Syndesmosis fixation: Analysis of shear stress via axial load on 3.5mm and 4.5mm quadricortical syndesmotic screws. *J Foot Ankle Surg.* 2006;45(2):65-69. **Citations = 132*
5. **Meyer EG**, Haut RC. Excessive compression of the human tibio-femoral joint causes ACL rupture before bone fracture. *J Biomech.* 2005; 38(11):2311-2316. **Citations = 184*
4. Von Pfeil D, Dejardin LM, **Meyer EG**, Weerts R, Decamp C, Haut RC. Biomechanical comparison of an interlocking nail and a plate-rod combination; An in vitro analysis in a canine fracture model. *Am J Vet Res.* 2005; 66(9):1536-1543. **Citations = 67*
3. Kirsch J, Dejardin LM, **Meyer EG**, Decamp C, Haut RC. Effect of Intramedullary Pin-plate combination on the mechanical properties of pantarsal arthrodesis; A comparative in vitro analysis in dogs. *Am J Vet Res.* 2005; 66(1): 125-131. **Citations = 21*
2. **Meyer EG**, Sinnott MT, Jayaraman GS, Smith WE, Haut RC. The effect of biaxial impact on the 90° flexed human knee joint stability and injury tolerance. *Stapp Car Crash Conf J.* 2004; 48: 53-70. **Citations = 10*
1. **Meyer EG**, Haut RC. The effect of knee impact angle on tolerance to rigid impacts. *Stapp Car Crash Conf J.* 2003; 74: 1-20. **Citations = 38*

FORMAL PRESENTATIONS

1. Explainable AI for Educators: Engaging Students and Inspiring Entrepreneurial Mindsets. KEEN National Conference. January 24, 2026: Oklahoma City, OK.
2. Applications of AI for Engineering, Science and Writing. LTU Assessment Day. September 16, 2025: Southfield, MI (Invited)
3. Tasting Science Seminar. MSU Pharmacology and Toxicology Club. September 16, 2025: East Lansing, MI (Invited)
4. Opportunities in Biotechnology: From medical devices to wearable technologies. Industrial Engineering Research Seminar March 19, 2025. University of Michigan Dearborn, MI. (Invited)
5. Using Simulation-Based Active Learning (SAL) and Course-Based Research Experiences (CRE) to motivate student engagement, technical concepts, laboratory skills and teamwork. LTU Research Day April 2024. Southfield, MI.
6. Driver Biomechanics & Monitoring. Ford Motor Co. Cabin Systems Research & Development Group Seminar. July 25, 2023: Dearborn, MI. (Invited)
7. Disruptive Biotechnology: Past and Future Promises of Biomedical Engineering and Wearables. LTU Research Day Presidential Colloquium. April 21, 2023: Southfield, MI. (Invited)
8. AI and the Future of Education. LTU Humanities + Technology. April 6, 2023: Southfield, MI.
9. Active Learning Modules for Training Engineers to Develop Innovative Orthopedics Applications. Summer Biomech, Bioeng, Biotrans Conf. June, 2020: Virtual.

10. Using "Industry 4.0" Technologies to Reinforce Course Topics Through Hands-on Skill Modules. LTU CTL: Lifelong Learners by Example; Technology Tools Across Campus. January 24, 2020: Southfield, MI.
11. Wearable Technology Products. LTU Centrepolis Accelerator Workshop: Developing Intelligent Connected Products: How to Make Your Hardware Product Smart! October 15, 2019: Southfield, MI. (Invited)
12. Sports Injury Prevention: How Industry 4.0 Advancements in Wearable Technology Can Save ACLs. LTU Research Day. April 2019: Southfield, MI.
13. Wearable Technology Products. LTU Centrepolis Accelerator Workshop: Developing Intelligent Connected Products: How to Make Your Hardware Product Smart! March 20, 2019: Southfield, MI. (Invited)
14. Victor Frankenstein Today: Shelley's Novel and Contemporary Bioengineering. LTU Humanities + Technology. October 11, 2018: Southfield, MI.
15. Building Interactive Wearable Technology Devices to Motivate Hands-on Experience with Biomechanics and Biomedical Engineering Design. World Congress of Biomechanics; July 11, 2018: Dublin, Ireland.
16. Hands-on activities for Prototyping and Ideation in the Wearable Technology Ecosystem. Detroit Public School Teacher Workshop; June 1-2, 2018: Southfield, MI. (Invited)
17. Innovation in Biomedical Engineering: Incorporating "Entrepreneurship" and "Maker" Mindsets. A2H2 Meetup; April 26, 2018: Ann Arbor, MI. (Invited)
18. Using a Reduced-order Dynamical Model to Determine the Effect of Posture on Knee Loading During Jump Landing. LTU Research Day. Southfield, MI, April 2018.
19. Reinforcing Technical Topics through Hands-on EML Design Modules. KEEN Conference; January 5, 2018: Jacksonville, FL.
20. Hands-on activities for Prototyping and Ideation in Biomedical Engineering. ASEE Annual Conf. June 28, 2017; Columbus, OH.
21. Engineering Validation of a Novel Fluted Osteotome for Surgical Expansion of an Implant Site in Bone. Osseodensification World Symposium: January 12, 2017; Orlando, FL (Invited).
22. Sports Medicine Injury Prevention: An Engineering Approach. Henry Ford Hospital Bone & Joint Center Seminar. September 30, 2016; Detroit, MI (Invited).
23. Determination of Ligament Strain During High Ankle Sprains due to Excessive External Foot Rotation in Sports. Providence Hospital Podiatry Residency Seminar. September 13, 2016; Southfield, MI (Invited).
24. Implementation of an Entrepreneurial Minded Learning Module Based on "Quantified Self" to Motivate Engineering Design in an Introductory Biomedical Engineering Course. ASEE NSC Annual Conf. March 19, 2016; Mount Pleasant, MI.
25. Learning Modules Focused on Entrepreneurship in Biomedical Engineering. ASEE NCS Annual Conf; March 18, 2016; Mount Pleasant, MI.
26. Using Quantified Self for EML in Biomedical Engineering Courses. QS4EML Roadshow: Ohio Northern University; March 11, 2016; Ada, OH (Invited).
27. Bone Biomechanics: Effects on Implant Osseointegration. Implants and Osseodensification; January 15, 2016; Jackson, MI (Invited)
28. Using Quantified Self for EML in Introduction to Biomedical Engineering. QS4EML Roadshow: Bucknell University; November 6, 2015; Lewisburg, PA (Invited).
29. Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the "Quantified Self" social movement. ASEE Annual Conference; June 6, 2015; Seattle, WA.
30. Quantified Self for Entrepreneurial Minded Learning (QS4EML) Webinar, Kern Family Foundation Engineering Entrepreneurship Network; May 7, 2015 (Invited).
31. Using Quantified Self for EML in a freshman level course. QS4EML Roadshow: Lawrence Technological University; April 24, 2014; Southfield, MI (Invited).
32. A novel osseodensification technique increases implant primary stability by compaction and auto-grafting bone. LTU Research Day. April 10, 2015. Southfield, MI.
33. Entrepreneurship at LTU. Innovating the Curriculum with Entrepreneurship in Biomedical Engineering; March 27, 2015: Southfield, MI (Invited).
34. General treatment for musculoskeletal problems and specifics of knee and lower extremity problems. Michigan State University College of Human Medicine: HM 517. October 14, 2014. East Lansing, MI (Invited).
35. Sports Medicine Injury Prevention: An Engineering Approach. Wayne State University Biomedical Engineering Anniversary Symposium. August 14, 2014. Detroit, MI (Invited).
36. Medical Leaps & Bounds: Thumb Education & Community Day Conference. May 9, 2014. Thumbworks! Marlette, MI (Invited).

37. Valgus Bending of the Knee Reduces the Tibiofemoral Compression Required to Cause ACL Injury. LTU Research Day. April 4, 2014. Southfield, MI.
38. Characterization of occupant lower extremity behaviour during moderate-to-high speed rear impacts. SAE World Congress. April 16, 2013. Detroit, MI. Also presented at LTU Research Day. March 29, 2013. Southfield, MI.
39. Sports Medicine Injury Prevention: An Engineering Approach. NFL Foot and Ankle Committee Meeting. November 17, 2012. Charlottesville, VA (Invited).
40. Determination of ankle ligament strain using a rigid body computational model for sports injury scenarios. IRCOB Conf. September 12, 2012. Dublin, Ireland.
41. The effect of cyclic hydrostatic pressure on the functional development of cartilaginous tissues engineered using bone marrow derived mesenchymal stem cells. *ASME Summer Bioengineering Conf.* June 25, 2011. Famington, PA.
42. Sports Medicine Injury Prevention: An Engineering Approach. OSU Sports Medicine. June 22, 2011. Columbus, OH (Invited).
43. Using cadaver knee simulation models to study non-contact ACL injury mechanisms. *IOC World Conference on Prevention of Injury & Illness in Sport.* April 8, 2011. Monte-Carlo, Monaco (Invited).
44. Low oxygen tension is a more potent regulator of chondrogenic differentiation than dynamic compression. *17th Congress of the European Society of Biomechanics*, July 7, 2010. Edinburgh, UK.
45. The effect of cyclic hydrostatic pressure and low oxygen tension on the chondrogenic differentiation of bone marrow derived mesenchymal stem cells. *8th International Conference on Cell and Stem Cell Engineering*, June 10, 2010. Dublin, Ireland.
46. Rotational injuries of the lower extremity and the use of an *in situ* risk assessment tool for rotational traction. *2nd International SportSURF STARSS Conference.* April 22, 2010. Loughborough, UK.
47. Biomechanical response of the knee to injury level forces in sports loading scenarios. *ACL Research Retreat V.* March 27, 2010. Greensboro, NC, USA.
48. Low oxygen tension: A more potent chondrogenic stimulus than dynamic compression for bone marrow derived MSCs in agarose hydrogel culture. RCSI/TCD Tissue Engineering Meeting. Oct. 7, 2009. Dublin, Ireland (Invited).
49. Computational and Experimental Mechanobiology: Fundamental advances and implementation for medical device engineering. Irish Medical Device Ass./SFI Forum. Oct. 1, 2009 Galway, Ireland (Invited).
50. Low oxygen tension is a more potent regulator of chondrogenic differentiation than dynamic compression. *Bioengineering in Ireland.* Jan. 22, 2010. Malahide, Ireland.
51. Biomechanical Response of the Lower Extremity to Injury Level Forces in Sports Loading Scenarios. Trinity Centre for Bioengineering Seminar Series. April 4, 2009, Dublin, Ireland (Invited).
52. Knee joint relative motion during ACL rupture by internal tibial torsion of tibiofemoral compression. *NACOB.* Aug. 9, 2008, Ann Arbor, MI.
53. Tibiofemoral contact pressures and osteochondral microtrauma from ACL rupture via hyperextension and joint compression. *NACOB.* Aug. 6, 2008 Ann Arbor, MI.
54. Correlating experimental joint motion with lower extremity ligament strain. *Biomechanics Seminar Series*, June 10, 2008, East Lansing, MI (Invited).
55. Anterior Cruciate Rupture Due to Excessive Internal Torque of the Human Tibia. *American Society of Biomechanics*, Aug. 22, 2007, Palo Alto, California.
56. An Experimental Model and In Situ Risk Assessment Tool for the Study of "High Ankle" Sprains, *COM Research Group*, January 11, 2007, East Lansing, MI (Invited).
57. High Ankle Sprains. *Sports Medicine, 2006*; MSU College of Osteopathic Medicine, May 12, 2006, East Lansing, MI (Invited).
58. The Effect of Biaxial Impact on the 90° Flexed Human Knee Joint Stability and Injury Tolerance. *48th Stapp Car Crash Conference*, November 1, 2004, Nashville, TN.
59. Running Shoe Testing: What It Means, Why It's Important. *Athletic Trainers Lecture Series*, October 28, 2004, East Lansing, MI (Invited).
60. The Effect of Knee Impact Angle on Tolerance to Rigid Impacts. *47th Stapp Car Crash Conference*, October 27, 2003, San Diego, CA.

ABSTRACTS (PEER-REVIEWED)

1. Li Y, Peponis J, Jiang H, He G, **Meyer EG**. Essential Skills in Biomedical Engineering. ASEE (In Press).

2. Choudhari A, **Meyer EG**. Magic Mirror: Design of a Computer Vision-Based Biomechanical Interface. BMES Conf. San Diego, CA, 2025.
3. **Meyer EG**. Sustainability Framework Applied to Biomechanics Course Modules to Motivate Opportunities and Collaboration. Am Soc Biomech Conf. Pittsburgh, PA, 2025.
4. Carpenter C, Smythe A, Huschke T, Dean H, **Meyer EG**. Impact360 Collar for Head Impact Risk Assessment. Am Soc Biomech Conf. Pittsburgh, PA, 2025.
5. Scagnetti D, Fuhrnamm J, Robinson S, Drvodelic A, **Meyer EG**. Design of a Concussion Recovery Device. Am Soc Biomech Conf. Pittsburgh, PA, 2025.
6. Wash A, Meyer EG. Integrating Wearable IMU Sensors to Collect and Analyze Human Movement Outside the Lab. Am Soc Biomech Conf. Knoxville, TN, 2023.
7. Rossman S, **Meyer EG**, Isaza J, Rundell S. The Risk of Recurrent Disc Herniation Following Decompression Surgery with Surgical Detachment of the Posterior Extensor Muscles. Summer Biomech, Bioeng, Biotrans Conf. Eastern Shore, MD, 2022.
8. Rossman S, **Meyer EG**, Isaza J, Rundell S. Physiological Loading Conditions to Simulate Standing. Summer Biomech, Bioeng, Biotrans Conf. Virtual, 2021.
9. Wieland B, Balint B, Jaramillo J, Barry D, Pan-Zagorski W, **Meyer EG**. Shear Pressure Vibration Comfort System. Summer Biomech, Bioeng, Biotrans Conf. Virtual, 2021.
10. Rossman S, **Meyer EG**, Rundell S. The Importance of Posterior Muscle Strength and Facet Contact in Establishing Sagittal Balance During Standing and Forward Bending. Am Soc Biomech Conf. Virtual, 2020.
11. Rossman S, **Meyer EG**, Rundell S. The Importance of Posterior Muscle Strength and Facet Contact in Reducing Disc Herniation Risk During Standing and Forward Bending. Am Soc Biomech Conf. Virtual, 2020.
12. **Meyer EG**, Kaminski BD. Active Learning Modules for Training Engineers to Develop Innovative Orthopedics Applications. Summer Biomech, Bioeng, Biotrans Conf. Virtual, 2020.
13. Rossman S, **Meyer EG**, Rundell S. Development of a Finite Element Lumbar Spine Model to Predict Intervertebral Disc Herniation Risk. ISB Conference; Calgary, Canada, 2019.
14. Sacdalan J, Rea E, **Meyer EG**, Yu N. Simple Human Gait Analysis: Model and Experiment. Michigan Academy of Science, Arts and Letters; Alma, MI, 2019.
15. **Meyer EG**, Verjdani HR. Using a Reduced-order Dynamical Model to Determine the Effect of Posture on Knee Loading During Jump Landing. World Congress of Biomech, Dublin, Ireland, 2018.
16. **Meyer EG**, Nasir M. Practicing Design and Prototyping in Biomedical Engineering Programs. BMES Conference; Phoenix, AZ, 2017.
17. Nasir, M, **Meyer EG**. Fostering the Entrepreneurial Mindset in Biomedical Engineering. BMES Conference, Minneapolis, MN, 2016.
18. **Meyer EG**, Nasir M. Student Learning of Biomechanics Topics By Embedding “Quantified Self” Motivated Problem Based Learning Modules In Biomedical Engineering Courses. Summer Biomech, Bioeng, Biotrans Conf. National Harbor, MD, 2016.
19. Weaver BT, Davison M, Rundell S, **Meyer EG**. Determination of Femur Fracture Pattern When Run Over By Moving Railroad Equipment. Summer Biomech, Bioeng, Biotrans Conf. National Harbor, MD, 2016.
20. Mozurkewich KR, Colarossi N, Issa M, **Meyer EG**, Nasir M. Fold-And-Go Single Knee Scooter. Summer Biomech, Bioeng, Biotrans Conf. National Harbor, MD, 2016.
21. **Meyer EG**, Nasir M. Implementation of an Entrepreneurial Minded Learning Module Based on “Quantified Self” to Motivate Engineering Design in an Introductory Biomedical Engineering Course. ASEE NSC Annual Conf. Mount Pleasant, MI, 2016.
22. Nasir M, **Meyer EG**, Li Y. Providing Opportunities for Applied Research and Design-Based Capstone Projects in Biomedical Engineering. ASEE Annual Conf. Seattle, WA, 2015.
23. Li Y, **Meyer EG**, Nasir M, Jensen E. Developing Key Performance Indicators for Effective Assessment of Student Outcomes in Biomedical Engineering Program. ABET Symposium. Atlanta, GA, 2015.
24. **Meyer EG**, Greenshields D, Huwais S. Increased Bone-Implant Contact and Bone Mineral Density with a Novel Osteotomy Preparation Technique Termed Osseodensification. Academy of Osseointegration. San Francisco, CA 2015.
25. Nasir M, **Meyer EG**. Entrepreneurship in small doses. VentureWell Open Conference, Washington, D.C., 2015

26. **Meyer EG**, Greenshields D, Huwais S. A Novel Osseodensification Technique Increases Implant Primary Stability by Compaction and Auto-Grafting Bone. American Academy of Implant Dentistry. Orlando, FL, 2014.
27. Petku L, **Meyer EG**, Knapp J. Finite Element Analysis Comparison of Two Types of Removable Partial Denture Designs. BMES Annual Conf. San Antonio, TX, 2014.
28. Greenshields D, Porter R, Killewald J, **Meyer EG**. Novel Design of an Anterior Cruciate Ligament (ACL) Injury Prevention Brace. BMES Annual Conf. San Antonio, TX, 2014.
29. **Meyer EG**, Greenshields D, Huwais S. Osseodensification Is A Novel Implant Preparation Technique That Increases Implant Primary Stability By Compaction And Auto-Grafting Bone. American Academy of Periodontology. San Francisco, CA, 2014.
30. **Meyer EG**, Wei F, Shimokochi Y, Haut RC. Valgus Bending of the Knee Reduces the Tibiofemoral Compression Required to Cause ACL Injury. World Congress of Biomechanics. Boston, MA, 2014.
31. Wei F, **Meyer EG**, Braman JE, Powell JW, Haut RC. The rotational stiffness of football shoes may affect the location of a potential ankle injury. ASB Ann Meeting. Gainesville, FL, 2012.
32. Button KT, Wei F, **Meyer EG**, Fitzsimons K, Haut RC. Determination of in situ ankle ligament strains in cases of high and medial ankle sprains. ASME Summer Bioengineering Conf. Farjardo, Puerto Rico, 2012.
33. Shimokochi Y, Igawa T, Oishi A, Kato N, **Meyer EG**, Ambegaonkar JP. Increased trunk stability modulates sagittal plane lower extremity kinetics and kinematics during a single-leg landing. Europe Soc of Sports Trauma, Knee Surg and Arthroscopy Cong. Geneva, Switzerland, 2012
34. **Meyer EG**, Buckley CT, Kelly DJ. The effect of cyclic hydrostatic pressure on the functional development of cartilaginous tissues engineered using bone marrow derived mesenchymal stem cells. ASME Summer Bioengineering Conf. Farnington, PA, 2011.
35. **Meyer EG**, Isaac DI, Haut Donahue TL, DeJardin LM, Haut RC. Comparisons of the joint response to surgical transection and traumatic rupture of the ACL in a rabbit model. ASME Summer Bioengineering Conf. Farnington, PA, 2011.
36. Wei F, Braman JE, **Meyer EG**, Powell JW, Haut RC. Mechanism of injury in a high ankle sprain: A simulation study. ASME Summer Bioengineering Conf. Farnington, PA, 2011.
37. Post JM, Wei F, Braman JE, **Meyer EG**, Powell JW, Haut RC. Mechanism of injury in a high ankle sprain: A cadaveric study. Michigan Osteopathic Association (MOA) Annual Convention. Dearborn, MI, 2011.
Scientific Research Award - Honorable Mention
38. Shimokochi Y, Ambegaonkar JP, **Meyer EG**. Influence of sagittal plane trunk positions on anterior tibial shear force during landings. National Athletic Trainers Association Annual Meeting, New Orleans, LA, 2011.
39. Shimokochi Y, **Meyer EG**. Sagittal plane body positions influence tibial anterior shear force during single-leg landing. IOC World Conference on Prevention of Injury & Illness in Sport. Monte-Carlo, Monaco, 2011.
40. **Meyer EG**, Buckley CT, Kelly DJ. The functional properties of cartilaginous tissues using bone-marrow derived MSCs improves as they are scaled towards geometries of clinical relevance. Orthop Res Soc Meeting, Long Beach, CA, 2011.
41. Buckley CT, **Meyer EG**, Kelly DJ. Expansion in the Presence of FGF-2 Modulates the Temporal Development of Cartilaginous Tissues Engineered Using Infrapatellar Fat Pad Derived MSCs. ORS Meeting, Long Beach, CA, 2011.
42. Haugh MG, **Meyer EG**, Thorpe SD, Vinardell T, Duffy G, Kelly DJ. Temporal and spatial changes in chondrogenic gene expression of MSCs in response to dynamic loading. ORS Meeting, Long Beach, CA, 2011.
43. Sheehy EJ, Vinardell T, **Meyer EG**, Buckley CT, Kelly DJ. The oxygen environment regulates mesenchymal stem cell expansion and chondrogenic differentiation in an in vitro cartilage defect model. ORS Meeting, Long Beach, CA, 2011.
44. **Meyer EG**, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension is a more potent regulator of chondrogenic differentiation than dynamic compression. European Society of Biomech, Edinburgh, UK, 2010.
45. **Meyer EG**, Buckley CT, Kelly DJ. The effect of cyclic hydrostatic pressure and low oxygen tension on the chondrogenic differentiation of bone marrow derived mesenchymal stem cells. Int Conf on Cell and Stem Cell Eng, Dublin, Ireland, 2010.

46. **Meyer EG**, Buckley CT, Haugh MG, Kelly DJ. The influence of expansion on the resulting functional properties of cartilaginous tissues generated from MSCs. UK MSC Meeting, Leeds, UK, 2010.
47. **Meyer EG**, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension is a more potent regulator of chondrogenic differentiation than dynamic compression for bone marrow derived mesenchymal stem cells. Tissue Eng and Regenerative Med Int Society, Galway, Ireland, 2010.
48. **Meyer EG**, Haut RC. Biomechanical response of the knee to injury level forces in sports loading scenarios. ACL Res Retreat V, Greensboro, NC, 2010.
49. **Meyer EG**, Haugh MG, Buckley CT, Kelly DJ. Monolayer expansion and cell seeding density influence the functional properties of cartilaginous tissues engineered using MSCs. Orthop Res Society Meeting, New Orleans, LA, 2010.
50. **Meyer EG**, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension: A more potent chondrogenic stimulus than dynamic compression for bone marrow derived MSCs in agarose hydrogel culture. Orthop Res Society Meeting, New Orleans, LA, 2010.
51. Haugh MG, **Meyer EG**, Thorpe SD, Vinardell T, Kelly DJ. The effect of loading on chondrogenesis at different stages of stem cell differentiation. Bioengineering in Ireland, Malahide, Ireland, 2010.
52. **Meyer EG**, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension is a more potent regulator of chondrogenic differentiation than dynamic compression. Bioengineering in Ireland, Malahide, Ireland, 2010.
53. Villwock MR, **Meyer EG**, Powel JW, Haut RC. External rotation ankle injuries: Investigating ligamentous rupture. ASME Summer Bioengineering Conf., Lake Tahoe, CA, 2009.
54. **Meyer EG**, Baumer TG, Haut RC. Knee joint relative motion during ACL rupture by internal tibial torsion of tibiofemoral compression. North Am. Congress on Biomechanics (NACOB), Ann Arbor, MI, 2008.
55. **Meyer EG**, Baumer TG, Haut RC. Tibiofemoral contact pressures and osteochondral microtrauma from ACL rupture via hyperextension and joint compression. NACOB, Ann Arbor, MI, 2008.
56. **Meyer EG**, Baumer TG, Haut RC. Tibiofemoral contact pressures and osteochondral microtrauma during ACL rupture due to excessive compressive loading and internal tibial torsion. NACOB, Ann Arbor, MI, 2008.
57. Villwock MR, **Meyer EG**, Powell JW, Fouty AJ, Haut RC. Football playing surface components may affect lower extremity injury risk. NACOB, Ann Arbor, MI, 2008.
58. Villwock MR, **Meyer EG**, Powell JW, Fouty AJ, Haut RC. Football shoe designs may affect lower extremity injury risk. NACOB, Ann Arbor, MI, 2008.
59. Isaac DI, **Meyer EG**, Kepich E, Haut RC. Acute chondrocyte damage and chronic changes in the rabbit tibiofemoral joint after impact. 54th Ann. Mtg. of the Orthopaedic Research Society, San Francisco, California, 2008.
60. **Meyer, EG**, Haut, RC. Anterior cruciate rupture due to excessive internal torque of the human tibia. Ann. Mtg. Am. Society of Biomechanics, Palo Alto, CA, 2007.
61. Guillou RP, Frank J, Dejardin LM, Olivier B, **Meyer EG**, Haut RC. Biomechanical evaluation of medial plating for pantarsal arthrodesis- A comparative in vitro study in dogs. Am. College Vet. Surg., 17th Ann Symp, Chicago, Illinois, 2007.
62. Guillou. RP, Cabassu, JB, Sinnott, MT, Ting, D, **Meyer, EG**, Haut, RC, Dejardin, LM. Effect of bending direction on the mechanical behavior of interlocking nail systems. Phi Zeta Research Day, College of Vet. Med., East Lansing, MI, 2007.
63. Guillou, RP, Frank, JD, Sinnott, MT, **Meyer, EG**, Haut, RC, Dejardin, LM. Biomechanical evaluation of medial plating for pantarsal arthrodesis—A comparative in vitro study in dogs. 17th Ann. Phi Zeta Research Day, College of Vet. Med., East Lansing, MI 2007.
64. Guillou, RP, Frank, JD, Sinnott, MT, **Meyer, EG**, Haut, RC, Dejardin, LM. Biomechanical evaluation of medial plating for pantarsal arthrodesis—A comparative in vitro study in dogs. Ann. Mtg. Am College Vet Surg, 2006.
65. Guillou RP, Cabassu JB, Ting D, Sinnott MT, **Meyer EG**, Haut RC, Dejardin LM. Effect of bending direction on the mechanical behavior of interlocking nail systems. Ann. Mtg. Amer. Coll. Vet. Surg., 2006.
66. **Meyer EG**, Handzo D, Haut RC. Chronic Changes in the Rabbit Tibiofemoral Joint Following a Single Blunt Impact. Orthop Res Soc Meeting, Chicago, IL, 2006.
67. Bonawandt KA, Dejardin LM, **Meyer EG**, Haut RC. In Vitro Mechanical Evaluation of a New 3.5/2.7mm Hybrid Plate For Pantarsal Arthrodesis in Dogs. ACVS Mtg., 2005.

68. Déjardin LM, Bonawandt K, **Meyer EG**, Lansdowne JL, Haut RC: In vitro mechanical evaluation of new 3.5/2.7mm hybrid plates for tarsal arthrodesis in dogs. 14th Annual European College of Veterinary Surgeons Scientific Meeting, Lyon, France, July 6-9, 2005, pp 379.
69. Von Pfeil D, Dejardin LM, **Meyer EG**, Weerts R, Decamp C, Haut RC. Biomechanical Comparison of an Interlocking Nail and a Plate-Rod Combination; An in vitro analysis in a canine fracture model. Vet Orthop Soc Meeting, 2004.
70. **Meyer EG**, Jayaraman V, Sevensma E, Haut RC. Anterior Cruciate Ligament Rupture Due to Compression of the Human Tibiofemoral Joint. Orthop Res Soc, 2003.
71. Kirsch J, Dejardin LM, **Meyer EG**, Decamp C, Haut RC. Mechanical evaluation of canine pantarsal arthrodesis. Am College Vet Surg, 2003.

GRANTS AND FELLOWSHIPS

- National Science Foundation S-STEM, In Review, 2026 - \$1,999,929, Role: Co-Primary Investigator. "Bridging the Gap and Building Future Engineers through Scholarships, Mentoring, Research, and Experiential STEM Learning for Low-Income Students."
- Lawrence Technological University Research Start-up Project, **Funded, 2026 - \$5,137** Role: Primary Investigator. "Early Biologic Treatment Approaches for Post-Traumatic Osteoarthritis."
- Agency for Healthcare Research and Quality (AHRQ), *Not Funded*, 2025 - \$960,864, Role: Co-Primary Investigator. "Violence Prediction and Mitigation using AI in Healthcare."
- Blue Cross Blue Shield of Michigan Foundation, *Not Funded*, 2025 - \$25,000, Role: Co-Founder. "Improving Hospital Outcomes Through Predictive Analytics."
- Michigan Health and Hospital Association Keystone Center, **Funded, 2024-2025 - \$125,000**, Role: Co-Founder. "Development of a Healthcare Worker Violence Reduction Center."
- Community Foundation for Southeast MI Capacity Building Grant, **Funded, 2024-2026 - \$70,000**, Role: Co-Founder. "Development of a Healthcare Worker Violence Reduction Center."
- Michigan Health Endowment Fund Grant, *Not Funded*, 2024-2026 - \$150,000, Role: Co-Founder. "Development of a Healthcare Worker Violence Reduction Center."
- State of Michigan FY 2022-2023 Appropriations, *Not Funded*, 2024 - \$100,000, Role: Co-Founder. "Funding for a grant to Lawrence Technological University to support the development of a Healthcare Worker Violence Reduction Center."
- Centropolis Accelerator, **Funded, 2023-2024 - \$14,875**, Role: Primary Investigator. "Quantitative Validation of Heads Up™ System for Technique Improvement with Short Term Training."
- Michigan Department of Transportation Requisition Proposal, *Not Funded*, 2023-2024 - \$385,041, Role: Co-Primary Investigator. "Safety Effectiveness of Non-Freeway Sinusoidal Shoulder Corrugations."
- State of Michigan FY 2023-2024 Appropriations, *Not Funded*, 2023 - \$850,000, Role: Director. "GF/GP one-time funding for a grant to Lawrence Technological University to support the development of a Wearable Technology Innovation Center."
- National Science Foundation Regional Engines: Type 1, *Not Funded*, 2022-2025 - \$999,972, Role: Primary investigator. "Wearable Technology Innovation Center (WTIC) to Promote Open Collaboration for Human Enhancement."
- Lear Corporation Industry Sponsored Project, **Funded, 2020-2023 - \$30,000**, Role: Primary Investigator. "SVP Comfort System."
- National Science Foundation National Robotics Initiative, *Not Funded*, 2022-2025 - \$747,300, Role: Co-Primary investigator. "Towards Autonomous Walking: Developing A Wearable Robotic Assistant for Pedestrians (WRAP) to Improve Safety and Efficiency in Workplace Environments."
- Centropolis Accelerator, **Funded, 2020-2021 - \$5,000**, Role: Primary Investigator. "BALListik Smart Basketball Shooting Glove POC Prototype."
- National Science Foundation Major Research Instrumentation, **Funded, 2019-2020 - \$72,142**, Role: Co-Primary investigator. "Acquisition of Eye Tracking Equipment to Support Behavioral Research at LTU."
- Kern Family Foundation KEEN Institutional Grant Subaward, **Funded, 2017 - \$20,000**, Role: Co-Primary investigator. "Interdisciplinary Wearable Technology Studio for Design and Entrepreneurship".
- Kern Family Foundation KEEN Topical Grant, **Funded, 2016 - \$25,000**, Role: Co-Primary investigator. "Creating a Successful Model for Dissemination and Adoption of Entrepreneurial Mindset in Biomedical Engineering".

- DENSO North America Foundation, **Funded, 2016 - \$50,000**, Role: Co-investigator. "Advanced Haptics for Automation and Control Systems"
- Lawrence Technological University Faculty Seed Grant, **Funded, 2015 - \$5,000**, Role: Primary investigator. "Adaptable Low Cost 3D Ultrasound Scanning System: Position Tracking"
- Kern Family Foundation KEEN Topical Grant, *Not funded, 2015 - \$40,000*, Role: Co-Primary investigator. "Interdisciplinary Wearable Sensor Studio for Design and Entrepreneurship".
- Microsoft HoloLens Academic Research Grant, *Not funded, 2015 - \$135,000*, Role: Contributing Faculty. "Revolutionizing Technology-Based Pedagogy and Education Across Academic Disciplines Lawrence Technological University".
- Venture Well Course and Program Grant, (Resubmission), *Not funded, 2015*, Role Co-Investigator. "Innovating for Medical Needs".
- Kern Family Foundation KEEN Topical Network Grant, **Funded, 2014 - \$20,000**, Role: Co-PI. "Building a Network for Faculty Collaboration on Entrepreneurship by means of a "Quantified-Self Roadshow"
- Venture Well Course and Program Grant, *Not Funded, 2014*, Role Co-Investigator. "Innovating for Medical Needs".
- Kern Family Foundation KEEN Topical Grant, **Funded, 2013 - \$40,000**, Role: Primary investigator. "Fostering the entrepreneurial mindset through the development of multidisciplinary engineering learning modules based on the "Quantified Self" social movement".
- Kern Family Foundation KEEN Topical Grant, **Funded, 2013 - \$27,400**, Role: Contributing Faculty. "Opportunity seeking in a highly regulated product sector: Medical device products from concept to investor pitch".
- DENSO North America Foundation, **Funded 2013 - \$50,000**, Role: Primary investigator. "Sensing, Perception and Control for the Development of Next Generation Robotic Systems"
- National Institutes of Health R-15, *Not funded, 2013*, Role: Primary investigator. "Mesenchymal stem cell targeting as an early treatment for post-traumatic osteoarthritis"
- NSF Major Research Instrumentation, *Not Funded, 2013*, Role: Co-investigator. "Biorobotics Laboratory – Simulation and control of humanoid behavior in medical robotics engineering"
- National Football League Charities Medical Research Grant, *Not Funded, 2012*, Role: Primary investigator. "Validation of a lower extremity model to determine *in vivo* ligament strains during football injuries."
- National Football League Charities Medical Research Grant, *Not Funded, 2012*, Role: Co-investigator. "Ankle motion criteria producing "high ankle" sprains in the laboratory and on the field."
- Department of Defense Congressionally Directed Medical Research Program; Discovery Award, 2012 - *Not Funded*, Role: Co-investigator. "Mesenchymal stem cell targeting as an early treatment for post-traumatic osteoarthritis"
- DENSO North America Foundation, **Funded, 2011 - \$50,000**, Role: Co-investigator. "Robotics program: Creation of a state-of-the-art laboratory in the general area of navigation and vehicle control"
- Lawrence Technological University Faculty Seed Grant, **Funded, 2011 - \$3,000**, Role: Primary investigator. "Anterior cruciate ligament microdamage induced by single and cyclic subfailure loading"
- AO Foundation, Switzerland, Start-up Research Grant, 2011- *Not Funded*, Role: Primary investigator. "Determination of *in vivo* ligament strains during high ankle sprains in collegiate sports."
- Center for Identification Technology Research, 2011 - *Not Funded*, Role: Co-investigator. "Impact of Extraneous Weights on Static and Dynamic Soft Biometrics"
- Irish Research Council for Science Engineering and Technology EMPOWER Postdoctoral Fellowship, **Funded, 2009-2010 - \$145,000**, Role: Primary investigator. "Cartilage mechano-response to hydrostatic pressure during postnatal development"
- MSU Dissertation Completion Fellowship, **Funded, 2008 - \$6,000**.
- National Football League Charities Medical Research Grant, **Funded, 2005 - \$120,000**, Role: Co-investigator. "An Experimental Model and In Situ Risk Assessment Tool for the Study of "High Ankle" Sprains"
- AO VET Center, Switzerland, Research Grant, **Funded, 2004 - \$19,200**, Role: Co-investigator. "Biomechanical evaluation of pantarsal arthrodesis constructs stabilized with a dorsal 3.5mm dynamic compression plate or a medial 3.5/2.7mm hybrid plate. A comparative in vitro analysis in dogs"
- MSU Graduate Travel Scholarship, 2003.
- MSU Graduate Student Fellowship, 2002.
- IEEE Study Abroad Scholarship, 2001.

PATENTS/ INTELLECTUAL PROPERTY DISCLOSURES

- Radomski D, Radomski G, Lenhart R, Hagaman J, Meyer EG. Devices And Methods For Assessing Athletic Skills. 2020
- Greenshields D, Meyer EG, Cook K. "An anterior cruciate ligament injury prevention brace hinge". 2014
- Grattan P, Jayaraman G, Haut R, Meyer E. Improved automotive knee bolster (IAKB) "Concave Knee Bolster". 2005

MANUSCRIPT REVIEWER

- Acta Biomaterialia
- American Society for Engineering Education
- Annals of Biomedical Engineering
- Biomechanics and Modeling in Mechanobiology
- Cell & Tissue Research
- European Cells and Materials
- Journal of Biomechanics
- Journal of Biomechanical Engineering
- Journal of Mechanical Behavior of Biomedical Materials
- Journal of Mechanics in Medicine and Biology
- Journal of Orthopaedic Research
- Journal of Sports Engineering and Technology
- Osteoarthritis and Cartilage
- PLOS One
- Tissue Engineering

MEMBERSHIPS

- American Society of Biomechanics
- Sigma Xi (Scientific Research Honor Society)
- Alpha Eta Mu Beta (Biomedical Engineering Honor Society)
- Biomedical Engineering Society
- European Society of Biomechanics
- American Society of Mechanical Engineering
- Tau Beta Pi (Engineering Honor Society)

SERVICE ACTIVITIES

- Engineering Unleashed Ambassador for the American Society of Biomechanics (2025-2026)
- Coach, KEEN Integrating Curriculum with Entrepreneurial Mindset 1.0 (2023-2026)
- Community Catalyst, KEEN (2023-2024)
- Representative, KEEN Leaders (2023)
- External Member, Institutional Review Board, Explico Engineering Co (2021-present)
- Member, ABET Task Force, LTU College of Engineering (2020-2023)
- Panelist, NSF Merit Review (2019, 2020, 2021, 2024, 2025)
- Chair, LTU Graduate Council (2019-2021)
- Representative, KEEN Panel, Milwaukee, WI (2019, 2022)
- Member; Tenure Review Subcommittee, College of Arts & Sciences (2018, 2022, 2023)
- Representative, Biomedical Engineering Advisory Board, LTU College of Engineering (2017-present)
- Member, LTU Graduate Council (2017-2021)
- Representative, University Assessment Committee (2017-present)
- Member, LTU Marburger STEM Center (2017 – present)
- Member; Tenure Review Subcommittee, Engineering Faculty Council (2017, 2019, 2022-present)
- Advisor, Alpha Eta Mu Beta LTU Chapter Student Group (2015-present)
- Textbook Reviewer, Biomedical Engineering, Springer Science+Buisness Media (2015, 2020, 2021)
- Member, Lawrence Tech Football Task Force, LTU (2015)

- Member, Campus Affairs Committee, LTU Faculty Senate (2015-2017)
- Member, Strategic Planning Sub-Committee on Research/Creative Work, LTU (2014)
- Reviewer, Masters Student Poster Competition, World Congress of Biomechanics (2014)
- Coordinator, Senior Projects Poster Competition, LTU Life Sciences Advisory Board (2011-2014)
- Member; Engineering Faculty Council, LTU College of Engineering (2011-2014)
- Member; Academic Reputation Committee, LTU (2013-2014)
- Member; COE Fundamentals of Engineering Design Projects Task Force (2013-2019)
- Member; Biomedical Engineering Program Advisory Board, St. Clair College (2012-2015)
- Presenter; Sports Injury Prevention, LTU "Ask the Engineer" Program (2012-present)
- Activity Leader, BME Summer Camp, LTU College of Engineering (2011-present))
- Member; Institutional Animal Care and Use Committee (IACUC), LTU (2010-present)
- Activity Leader, Exploration Day, LTU Biomedical Engineering (2010-present)
- Member; Social Media Task Force, LTU (2011)
- Presenter; Blue Devil Day, LTU College of Engineering (2010, 2011, 2012)
- Member; Faculty Search Committee, LTU College of Engineering (2010, 2011, 2012, 2014, 2015, 2016, 2017, 2018)
- Representative, Life Science Advisory Board, LTU Colleges of Engineering and Arts & Science (2010-2016)

PROFESSIONAL WORKSHOPS/MEETINGS

- EPIIC Industry and Academia Summit: February 27, 2026: University of Detroit Mercy, MI.
- KEEN National Conference: Jan 22-24, 2026: Oklahoma City, OK.
- Reproducible Rehabilitation Research Summit: September 26-28, 2025: Los Angeles, CA.
- MichBio Medical Device Summit: Sept 25, 2025: Grand Rapids, MI.
- Workshop: Establishing University Partnerships with External Organizations NSF EPIIC Grant. May 15-16, 2025: Southfield, MI.
- MedHealth Summit: April 23, 2025: Detroit, MI.
- KEEN National Conference: Jan 23-25, 2025: Austin, TX
- Michigan Health & Hospitals Association Conference: June 26-28, 2024: Mackinaw Island, MI.
- KEEN Sustainability Workshop: June 3-6, 2024: Minneapolis, MN.
- MedHealth Summit: April 16, 2024: Detroit, MI.
- KEEN National Conference: Feb 8-10, 2024: Austin, TX
- KEEN ComCat Kickoff Workshop: September 20-21, 2023: Waukesha, WI.
- MichBio Life Sciences Showcase: September 18, 2023: Lansing, MI.
- KEEN MakerSpark Workshop: July 18-20, 2023: Boston, MA.
- Michigan Health & Hospitals Association Conference: June 29-30, 2023: Mackinaw Island, MI.
- NSF Technology Innovation & Partnerships, Use-Inspired Science Workshop: June 12-14, 2023: Boise State University, ID. Online Microlabs (April, May, Aug & Oct, 2023)
- Integr8 Conference: May 9, 2023: Novi, MI.
- KEEN National Conference: Jan 27-28, 2023: Atlanta, GA.
- MedHealth Summit: Nov. 1, 2022: Detroit, MI.
- Integr8 Conference: May 10, 2022: Novi, MI.
- ABET Symposium: April 14-16, 2021: Virtual Meeting.
- KEEN Leadership Unleashed Workshop: June 2-3, 2020: Indianapolis, IN (moved online due to Covid-19).
- Integr8 Conference: 2020: Virtual Meeting
- MedHealth Summit: April 18, 2019: Detroit, MI.
- ABET Symposium: April 11-12, 2019: Dallas, TX.
- MedDevice Forum: May 22, 2018: Grand Rapids, MI.
- KEEN/Bucknell University BFab Workshop: August 8-10: Lewisburg, PA.
- KEEN National Conference: Jan 3-5, 2018: Dallas, TX.
- Integr8 Conference: November 9, 2017: Detroit, MI.
- KEEN National Conference: Jan 4-6, 2017: Jacksonville, FL.
- Winter Meeting, KEEN: Jan 4-6, 2016: Tempe, AZ.

- Entrepreneurial Minded Learning, KERN Innovative Teaching Program; May 14-15, 2015: Southfield, MI.
- Winter Meeting, KEEN: Jan 5-8, 2015: Tempe, AZ.
- Winter Meeting, KEEN: Jan 3-4, 2014: Tempe, AZ.
- Fall Meeting, KEEN: September 26-28, 2013: Milwaukee, WI.
- Technology in Course Design and Instruction, LTU Summer Symposium; June 25-28, 2012: Southfield, MI
- Active and Collaborative Learning, KERN Innovative Teaching Program; May 21-24, 2012: Southfield, MI
- Entrepreneurially-Minded First Year Experience, KEEN; March 30, 2012: Flint, MI
- Exploring Ethical decision making in Engineering (E³), SEED; October 24, 2011: Ann Arbor, MI
- Integrated Course Design for Significant Learning, Dee Fink and Associates; August 24, 2011: Southfield, MI
- Problem Based Learning, KERN Innovative Teaching Program; May 16-19, 2011: Southfield, MI
- Shaping Entrepreneurial Engineers, KEEN; January 2-4, 2011: Tempe, AZ

GRADUATE STUDENT ADVISOR

- Andrew Ulaszek, Biomedical Engineering Design Project, LTU (expected 2026)
- Phil List, Biomedical Engineering Research Project, LTU (expected 2025)
- Ankita Choudhary, Biomedical Engineering Design Project, *LTU Research Day Poster Award* (MS, 2025)
- Carolin Minners, Biomedical Engineering Design Project, *LTU Research Day Poster Award* (MS, 2023)
- Nicole Carlin, Biomedical Engineering Design Project, LTU (MS, 2023)
- Vaishnevi Phutane, Biomedical Engineering Design Project, LTU (MS, 2023)
- Salvatore Brauer, Biomedical Engineering Design Project, LTU (MS, 2023)
- Anthony Wash, Biomedical Engineering Research Thesis, LTU (MS, 2022)
- Brian Czarnecki, Biomedical Engineering Research Thesis, LTU (MS)
- Stephanie Rossman, Mechanical Engineering Dissertation, LTU (PhD, 2021)
Biomechanics of Intervertebral Disc Herniation: *In Silico* Examinations of Spinal Decompression Surgery and Its Effect on Intervertebral Disc Herniation Risk
- Subathra Baskar, Biomedical Engineering Design Project, LTU (MS, 2020)
- Joseph Txarola, Biomedical Engineering Design Project, LTU (MS, 2020)
- Ahron Wayne, Biomedical Engineering Design Project, LTU (MS, 2020)
- Wesley Allen, Bioengineering Design Project, TCD (MSc, 2010)
- Johannes Schmid, Bioengineering Design Project, TCD (MSc, 2009)

GRADUATE STUDENT ADVISORY COMMITTEES

- Alexis VanBaarle, Forensic Anthropology PhD Committee, MSU (expected 2028)
- Justin Trzeciak, Mechanical Engineering Thesis, LTU (MS, 2019)
- Nicole Ravenscroft, Biomedical Engineering Thesis, LTU (MS, 2018)
- John Peponis, Biomedical Engineering Thesis, LTU (MS, 2018)
- Andrew Lyzen, Biomedical Engineering Thesis, LTU (MS, 2018)
- Keith Button, Mechanical Engineering Dissertation, MSU (PhD, 2015)
- Wei Yao, Mechanical Engineering Thesis, LTU (MS, 2012)

UNDERGRADUATE STUDENT RESEARCH

- Sarah Trapp and Leah Hall, BME Research – Designs and Methods for 3D Printed Prosthetic Hands (2019)
- Emily Gandolfi, LTU Presidential Research Award - Measuring the Effect of Posture on Lower Extremity Biomechanics During Jump Landing. (2019)
- Daniel Piotrowski, BME Student Assistant – Electronics and IOT Projects for Wearable Technology. (2019)
- Edward Rea, BME Research - Simple Human Gait Analysis: Model and Experiment. (2018)
- Hazem Elhusain Elnegres, BME Research - An Investigation into the Biomechanics of maximum allowable force on a gait mobility harness vs ceiling harness. (2018)
- Alejandra Farias Elizondo, BME Research - Computed Model of Lower Limb Joint Including Muscle Force, Ligament strain, and Joint Contact (2017)
- Daniel Moscoso, BME Research & President's Fellowship- Decoding hand gestures through EMG (2017)
- Zach Cowen, EBL Student Assistant – Adaptable Low Cost 3D Ultrasound Scanning System (2016)

- Brian Ziola, EBL Student Assistant – Adaptable Low Cost 3D Ultrasound Scanning System (2016)
- Aimee Tomlinson, EBL Student Assistant – Mechanical Testing of various medical devices (2016)
- Patrick Caren, EBL Student Assistant – Development of a bone plasticity FEM (2016)
- John Peponis, BME Research - The Juggling Cycle: An Investigation into the Biomechanics of Juggling (2016)
- Nicholas Denney, EBL Student Assistant – Development of Wearable Biomechanics Sensors (2015)
- Ryan Reed, EBL Student Assistant – Mechanical Testing Verification of ROMD Pedicle Screw System (2015)
- Colin Thompson, O'Malley Research Scholar Project – Quantification of Lower Extremity Kinematics During Martial Arts Roundhouse Kicking (2015)
- Dominique Reed, BME Research - Transcutaneous Stimulation of the Ulnar and Radial Nerve to Control Spasticity (2015)
- Kevin Mozurkewich, BME Research - Material Properties of Metal Products Produced Through Direct Metal Laser Sintering (2015)
- Benjamin Applebee, Computer Science Directed Study – Motion capture for video game design. (2015)
- Avery Weigle, Industrial Design Studio, Warrior Endurance 3D Printed MCL Protection Brace Spring (2015)
- Lindsay Petku, EBL Student Assistant - Finite Element Analysis Comparison of Two Types of Removable Partial Denture Designs (2014)
- Akram Alsamarae, EBL Student Assistant - Head Impact Testing: Hybrid III Dummy Head and Neck (2014)
- Dan Greenshields, BME Directed Study and EBL Student Assistant - Engineering validation of a novel "Osseodensification" preparation technique for surgical expansion of an implant site in bone. (2013-2014)
- Justin Killewald, LTU Quest Project – Three-dimensional range of motion measurement of total shoulder joint reconstruction with Latissimus Dorsi tendon transfer. (2013)
- John Schoenbeck - EBL Student Assistant Engineering validation of a novel fluted osteotome for surgical expansion of an implant site in bone (2013)
- Jeremy Quinlin, Computer Science Directed Study – Three-dimensional computer animation from human motion capture. (2012)
- Samantha Hutson, BME Directed Study - Development of a computational model of the human knee joint. (2012)
- Brian Figueiredo, BME Directed Study - Development of a computational model of the human lower extremity. (2011)

BME SENIOR PROJECTS

- Catelyn Chapin, Ethan Hochendoner, Joshua Lee, Donovan Sayyae. Customized shoulder bracing for athlete recovery. (2025-2026)
- Chloe Carpenter, Alaina Smythe, Taylor Huschke, Hannah Dean. Impact360 Collar for Head Impact Risk Assessment. *LTU Research Day Poster Award* (2024-2025)
- Daniela Scagnetti, Jeffry Fuhrmann, Sydney Robinson, Amber Drvodelic. Design of a Concussion Recovery Device. *Poster presented at Am. Soc. Biomechanics Ann. Conf. 2025, Pittsburgh, PA* (2024-2025)
- Sophia Buckberger, Keith Kolaczyk, Gabrielle Pryor. Healthcare Worker Violence Prevention (2023-2024)
- Lucas Adams, Adam Knapp, Samantha Nseir, Andrew Ulaszek. SPV Comfort System 3.0. Industry Sponsored Project by Lear (2022-2023)
- Sophia Judge, Bailey Wills, Sam Weber, Liz Beckmon. SPV Comfort System 2.0. Industry Sponsored Project by Lear (2021-2022)
- Brittany Wieland, Bethany Balint, Drake Barry, Juan Jaramillo. Comfort-Man: A quantitative method for long-term comfort assessment of automotive seating. Industry Sponsored Project by Lear (2020-2021)
**LTU BME Industry Advisory Board Best Poster Award*
- Alexis Snider, Evan Szabo. Knee Injury Prevention Wearable Device (2018-2019)
- Rania Anoni, Samantha Wheeler. Reducing Edema One Foot at a Time (2017-2018)
- Trevor Raymond, Brandon Schick. Formula Hybrid Racing Team: Tackling Driver Fatigue (2017-2018)
- Alejandra Farias, Justin McAllister. ApparOrtho: Total Knee Arthroplasty Balancing System (2017-2018)
- Ghislain Mammo, Edward Rea, Counter And Sink Extension (2017-2018)
- Alexandria Glumb, Fatimah Aljafer. Walk&Lift Cane (2016-2017)

- Kevin Ritchey, Norah Hammad. StretchPRO: A Leg Separation Apparatus (2016-2017)
- Kevin Mozurkewich, Nicholas Colarossi, Muntha Issa. Fold-and-Go Single Knee Scooter (2015-2016)
- Adam Gohl, John Peponis. Auto-TRAC for Carpal Tunnel Therapy (2015-2016)
- Patricia Nahas, Christopher Harris, Evan McKenzie. Bike Raptor: Upper Body Exercise Machine (2014-2015)
- Brandon Churches, Seth Klein. Reduction in Driver Lateral Shifting in the Blue Devil Motorsports SAE Hybrid Car. (2014-2015)
- Greenshields Daniel, Porter Rachel, Killewald Justin. Novel Design of an Anterior Cruciate Ligament (ACL) Injury Prevention Brace (2013-2014) **LTU Life Science Industry Advisory Board Best Poster Award*
**BMES 2014 Annual Conference Undergraduate Extended Abstract Award*
- Michael DiCicco, Stephen Osterhoff, Trevor Taormina. Implantable Endpad (2013-2014)
- Olesya Motovylyak, Michael Newton. High Resolution Imaging of the Intervertebral Disc Degeneration using EPIC-microCT (2012-2013) **LTU Life Science Industry Advisory Board Best Poster Award**
- Kevin Roberts, Katelyn Fortin. Heel Insert Senior Project (2011-2012)
- William Addis. Simulation of human femur bone fracture caused by a pedestrian-train accident (2011-2012).
- Brian Figueiredo, Joseph Leuenberger, William Slesinski. Balance Assist (2010-2011)
- Allison Andre, Danielle Beski, Erick Blank. Novel Bioreactor for Tissue Engineering (2010) **LTU Life Science Industry Advisory Board Best Poster Award*

OUTREACH CONTRIBUTIONS

- [KEEN Undergraduate Research](#) Excite and Engage Students to Explore Video: [What is Research?](#)

PERSONAL

Born: May 26, 1980 in Rochester, MI.

Graduated Brandon HS and OSMTech Charter School in Clarkston, MI: May, 1998.

Married to Susan. Children are Isabella, Nolan and Emmett.

Interests: family, travel, volleyball, soccer, golf, running, reading, music.